

Challenges and Barriers Facing Women in the IS Workforce: How Far Have We Come?

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ABSTRACT

In 2006, Riemenschneider et al. asked women working in the IS department at a Fortune 500 company what workplace barriers they faced that had influenced their voluntary turnover decisions that men did not. The current investigation offers a replication and extension of the Riemenschneider et al. (2006) study.

Specifically, this research begins to examine what changes have (or have not) taken place in the IS workplace regarding the challenges and barriers facing women in the IS field. Preliminary findings indicate that while *Barriers: Promotion, Managing Family Responsibilities, Work Schedule Flexibility, Work Stress* and *Turnover* were the key concepts identified by Riemenschneider et al. (2006), *Work Stress* is one of the few key issues that remains salient. What seems to be more salient for the current participants are issues related to the work environment, such as politics and bureaucracy (e.g., *Barriers: Politics*).

Categories and Subject Descriptors

K.7.1 Occupations; K.6.1 Project and People Management: Staffing; K.4.2 Social Issues: Employment

General Terms

Management

Keywords

IS Workforce, Gender, Barriers, Qualitative

1. INTRODUCTION

In December 2009, The Economist reported that women not only make up the majority of the professional workforce in many countries, but also that they earn nearly 60 percent of university degrees in America and Europe (Women in the Workforce: Female Power, 2009). Unfortunately, the statistics for women in the IS field are not so glowing. Only 18% of computer and information science degrees awarded in 2008 went to women, down from 37% in 1985, according to a study by the National

Center for Women & Information Technology, and about 56% of women in technical fields leave their jobs at the midlevel point, which is more than double the rate of men (Collett, 2010).

For almost a decade, researchers have sought to identify barriers to the retention of female IS employees (e.g., Ahuja, 2002; Trauth and Howcroft, 2006). Riemenschneider et al. (2006) studied female IS employees and identified key workplace barriers that often lead to voluntary turnover decisions (e.g., managing family responsibilities, stress, work schedule flexibility, job qualities, promotion barriers, discrimination and a lack of consistency on the part of management). The current investigation examines what changes have taken place regarding the barriers identified for women in the IS workplace four years later. Have the barriers facing women been removed? If so, why are women still underrepresented in the IS field and continue to leave the profession?

Both the 2006 study and the current research use the same qualitative methodological tool and an interpretivist perspective to identify respondents' cognitions regarding the perceived barriers. Consistent with the Riemenschneider et al. (2006) study, we asked women in the in-house IS department of one of the organizations used in the 2006 study to discuss what challenges they face that their male colleagues might not. What is reported here is the initial causal map evoked from the current respondents and a preliminary comparison to the previous work. A literature review is followed by a discussion of the method and then we conclude with our initial findings and an outline of the next steps.

2. LITERATURE REVIEW

In the first part of this longitudinal study, Riemenschneider et al. (2006) conducted a series of interviews aimed at uncovering women's perceptions of barriers they face in the IT field. The current study is a follow-up study to Riemenschneider et al. (2006) that aims to investigate perceptual changes, if any, which have occurred among women in the IT workforce.

In their initial literature review, Riemenschneider et al. (2006) focused on voluntary IT employee turnover, barriers in within the workplace, work-family conflict, and work stress. Since the original publishing, research on the topic of the barriers faced by women in IS has been limited. What has been published on the topic of gender and barriers/turnover in IS ranges from a discussion of parity in academia (Johnson et al., 2008; Lamp, 2007) to critical theory (Reid et al., 2010) to gender identity in the IS field (Adam et al., 2006; Woszczyński and Shade, 2010). With regard to retention and turnover scholars have utilized the

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individual differences theory of gender to explore retention (Trauth et al., 2009), and have examined differences in stress-related effects on turnover within the IS workforce such as burnout (Chilton et al., 2010; Chilton et al., 2005; Pawloski et al., 2007) and work overload (Ahuja et al., 2007) while addressing coping and adjustment among personnel in the ever-changing field of IS (Love and Irani, 2007).

Based on our review of the literature, it seems clear that the challenges facing women in the IS workforce is still a topic worthy of exploration. Women in IS continue to face glass ceilings and there is still no definitive explanation for the high levels of voluntary turnover. This research attempts to further clarify these issues by: 1) identifying the perceived workplace barriers women in IS experience; 2) identifying which barriers are related to turnover; and 3) determining the status of the barriers facing women in IS since first studied in 2006. The next section addresses the methodology we employed to explore the phenomenon.

3. METHOD

Using a social constructionist perspective (e.g., Berger and Luckmann, 1966; Dick and Nadin, 2006) and a qualitative data analysis method we seek to elucidate and graphically represent the emergent cognitions evoked from group interaction. From the cognitive representations we identify the unique challenges facing women in IS (content) and the organization of these challenges (structure) within the participants' mental models (Nadkarni and Narayanan, 2005).

What is presented here is part of a larger project that looks at IS workers' perceptions of challenges faced in the IS workplace. The researchers started investigating similar issues in 2002, and this study follows previous studies on similar topics. We conducted five focus groups with women working in the IS department at the headquarters of the same organization as previously explored. Participants were identified by the CIO and recruited via email invitation and the contact person within the organization handled the room scheduling and invitation process. The focus groups were held during working hours in on-site conference rooms. The organization, FoodCo (a pseudonym), is one of the world's largest processors and marketers of protein, and the second-largest food production company in the Fortune 500. The company has approximately 117,000 team members at more than 400 facilities and offices in the United States and around the world.

Our participants were 15 female IS workers, which, while small, is consistent with sample sizes in other qualitative studies (e.g., Gershon et al. 2004). Job titles of the participants included Business Analyst, Database Administrator, Organizational Change Specialist and Director of Business Process Change, to name a few. Due to the nature of some topics discussed in the larger study, and to increase interviewee trust in the researchers, a decision was made to gather limited demographic information from the participants.

The focus group interviews, which ranged from 45 to 65 minutes, were tape recorded and then transcribed verbatim. The responses to the question, "Do you think women in the IS workplace face different challenges than men? Please explain." provided the data for this study. To ensure confidentiality each participant was given a number, and asked to use those numbers instead of each other's names in the conversation. Two researchers were present for each focus group. The researchers were trained in group

interview techniques to promote consistency and reduce interviewer variance. While differences in the evoked mental models could be attributed to the different researchers, we do not believe this to be the case. The interview guide was strictly adhered to by the researchers as confirmed in the verbatim transcriptions. While steps were taken to mitigate researcher bias in the data collection and analysis processes, the researchers acknowledge that they are representing the concepts identified by the participants as experienced through their own subjectivity.

3.1 Data Analysis

In the data analysis phase our task was to elicit the relevant concepts from what participants said regarding the unique challenges women in IS profession face and map these concepts into a structural representation. We used revealed causal mapping (RCM), a multi-step process previously described in the literature (e.g., Narayanan and Armstrong, 2005; Riemenschneider et al., 2006) to access the salient concepts and cognitive connections between the concepts. This analytical technique has been applied in a wide variety of disciplines and has been applied in various ways to the IS discipline (e.g., Lee and Kwon, 2008; Nelson et al., 2000; Siau and Tan, 2005; Tegarden and Sheetz, 2003). Due to space limitations, much of the detail regarding the method has been omitted. Interested readers should refer to Riemenschneider et al. (2006), or Narayanan and Armstrong (2005) for details of the revealed causal mapping method. We provide an abbreviated description of the method here.

The four major steps in RCM are gathering the narratives from the focus groups, identifying the causal statements through the use of keywords (e.g., because, so, if-then) (Axelrod, 1976), separating the statements into the 'effect' and 'cause', and developing a coding scheme. A total of 132 causal statements were identified in the transcripts. In the coding process, frequently mentioned words in the statements are grouped together (Narayanan and Fahey, 1990), and a word or word group (i.e., concept label) summarizes the statements. The researchers assign the concept labels; however the concepts themselves emerge from the participants through the phrases captured in the language of the participants. As this study is a comparison to a previous study, the coding scheme developed in the first study was used. Any disagreements among the researchers regarding identification of a statement or the coding of those statements are resolved through discussion leading to a 100% level of agreement. The maps are drawn by replacing the participants' phrases with the concept labels.

Once the maps are drawn, measures for the analysis and comparison of the maps are developed: adjacency, which represents the strength of the direct linkage (relationship) between two concepts and contained values between 0 and 10, indicating the frequency (number) of mentions of this relationship; reachability, which includes both the direct and indirect effects of one concept on another; and centrality, which is the ratio of the direct linkages involving the concept divided by the total number of linkages in the map (Knoke and Kuklinski, 1982). While the adjacency matrix provides the foundation, the reachability matrix allows a more holistic picture of the causal relationships and thus the reachability numbers are reported on the line between the concepts on the causal map. Centrality measures how central or involved the concept is in the map, and is calculated as the ratio of the direct linkages involving the concept divided by the total number linkages in the map.

4. PRELIMINARY RESULTS

Per the social constructionist perspective, we do not claim to have *the* interpretation of the data, but *an* interpretation grounded in our understanding of the literature and the data. Descriptions of the key concepts that emerged are included in the Appendix.

4.1 Map Content and Structure

Figure 1a shows the aggregate map that reveals 31 concepts that the participants in the focus groups identified as the unique challenges facing women in the IS profession. Of the 31 concepts, the participants identified eight concepts specifically as barriers, and five concepts directly related to turnover. The remaining concepts identified were focused around the workplace environment and work-family balance issues. Recall that reachability is a measure of the strength of connection between two concepts, and the concepts shown on the map in Figure 1a are a result of using a reachability cutoff of >0.100 . This cutoff value allows a maximum inclusiveness of concepts within the constraint of map readability. The reachability is indicated by the number next to the arrow on the map, and takes into account all paths (both direct and indirect) from one concept to another. The concepts with the highest reachability (i.e., strongest relationship) are between the *Positive Comments* concept and the *Opportunities* concept (1.000), followed by *Negative Comments* and *Stress* (.405), *Work Stress* and *Negative Comments* (.401), and *Barriers: Women's Characteristics* and *Barriers: Politics* (.400) and *Barriers: Women's Characteristics* and *Barriers: Respect* (.400).

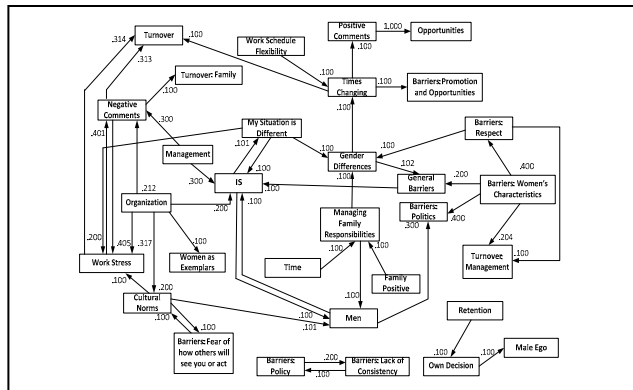


Figure 1a. Current Revealed Causal Map

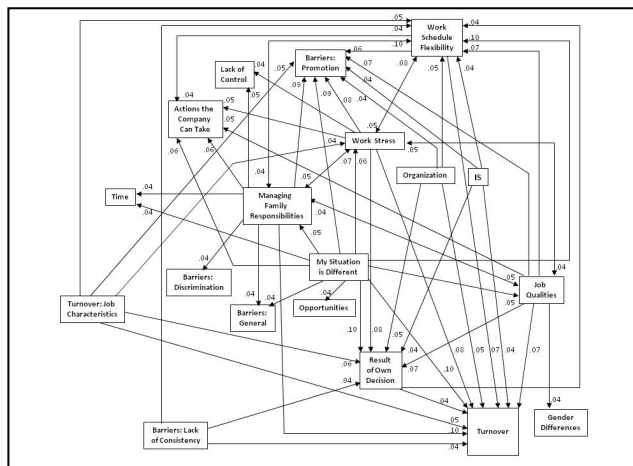


Figure 1b. Previous Revealed Causal Map

Some of the concepts are more central than others. Recall that a high centrality measure means that the concept is at the core of the mental model. The concepts possessing the highest centrality are *IS* (.778), then *Negative Comments* and *Work Stress* (.667), followed by *Cultural Norms*, *Men*, *Organization*, *Times Changing*, and *Gender Differences* (.556). The centrality data are presented in Table 1.

Table 1. Map Comparison Data

Measure	Previous Map	(a)	Current Map	(b)
Reachability	MFR → Turnover	.100	PC → Opportunities	1.000
	MSD → Turnover	.100	NC → Work Stress	.405
	MSD → ROD	.100	Work Stress → NC	.401
	MSD → WSF	.100	WC → Barriers: Politics	.400
	MFR → WSF	.100	WC → Barriers: Respect	.400
Centrality	Barriers: Promotion	.999	IS	.078
	Turnover	.940	Work Stress	.067
	Work Stress	.789	NC	.067
	MFR	.789	Cultural Norms	.056
	WSF	.752	Organization	.056
	Job Qualities	.752	Men	.056

MFR = Managing Family Responsibilities; MSD = My Situation is Different; ROD = Result of Own Decision; WSF = Work Schedule Flexibility; PC = Positive Comments; NC = Negative Comments; WC = Women's Characteristics

4.2 Initial Comparison & Next Steps

The map developed by Riemenschneider et al. (2006) is included as Figure 1b. There are 14 concepts in common for the two maps, 17 concepts appearing on the current map that did not appear previously, and there are five concepts on the previous map that were not articulated by the current respondents. Of the new concepts, five are focused on barriers faced by the women. There are only three relationships in common: between *Work Stress* and *Turnover*, between *Time* and *Managing Family Responsibilities*, and between *My Situation is Different* and *Work Stress*.

The reachability and centrality numbers are map/sample specific and cannot be directly compared. For example, centrality is the ratio of the concept linkages divided by the total linkages in the map. As the total number of linkages in the previous map was different than the number in the current map, the scale is different. The key to comparing the numbers is the relative order. Interestingly, when comparing the centrality numbers for the two groups, there is very little overlap. For example, *IS* was not mentioned on the previous map, and yet it is the most central concept in the new map. In contrast, *Work Stress* was in the second tier of centrality in the previous map, and remains in the second tier in the current map. Thus, it seems clear that *Work Stress* is the one constant across time.

When we combine the reachability and centrality measures we see that *Barriers: Promotion*, *Managing Family Responsibilities*,

Work Schedule Flexibility, Work Stress and Turnover were the key concepts identified by Riemenschneider et al. (2006). In contrast, while *Work Stress* remains a key issue, what seems to be more salient for the current participants are issues related to the organization, such as politics and bureaucracy (e.g., *Barriers: Politics*).

It is important to note that at the time of the data collection, FoodCo was engaged in a multi-pronged change process with initiatives ranging from expansions in production capacity to turnover within the Board of Directors and executive personnel. Also, during the time of data collection, FoodCo worked extensively on its Team Member Bill of Rights (TMBR) and statement of core values. Several aspects of FoodCo's TMBR that are relevant to our topic of study include the right to be free from discrimination, the right to compensation for work performed, and the right to continued training including supervisory training.

The authors are in the process of further analyzing and interpreting the current data collection, comparing the current findings to the previous findings, and exploring theories within which to frame the findings. We hope to further refine the initial model developed by Riemenschneider et al. (2006) and contribute to the practice of managing IS personnel.

5. ADDRESS INFORMATION

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7. APPENDIX

Brief descriptions of the core concepts that emerged from the participants with sample phrases captured in the language of the respondents are included here. The concept labels listed below were developed to capture the essence of the participants' phrases.

Actions the Company Can Take

Description: Actions the company can take to solve problems, increase performance, bring in new blood. Examples: "Your computer hook up from home to the mainframe," "Stagnant management holders are moved out"

Barriers: Discrimination

Description: Examples of people not being treated equally; statements about how things are not fair; discrimination in terms of the money they receive. Examples: "They don't see that we need to be paid for what we do for our position," "My increases stopped when I moved into IT"

Barriers: General

Description: Barriers they are experiencing in the workplace that do not fit into any of the more specific barriers. Examples: "You have more people in your group," "I don't live really close"

Barriers: Lack of Consistency

Description: Comments about a lack of consistency at the organizational level, a lack of consistency on the part of the supervisor in how treats employees, how some people in the department get treated a certain way and not others, how different

supervisors do different things. Examples: "Each manager feels a totally separate way," "it depends on the manager"

Barriers: Management

Description: Deals with upper management's attitudes, beliefs and actions which seem problematic for women; ineffective management. Examples: "Then there's also management, higher up that I deal with" "They're the ones that are approving the projects"

Barriers: Policy

Description: Lack of policies, or policies not followed. Examples: "There is no overarching policy" "Some people want to work part-time, but aren't allowed to cause it depends on your job"

Barriers: Politics

Description: Informal interaction with management; schmoozing; need to be political. Examples: "I wouldn't be able to go out to lunch with my team" "you're not going to cut it on the team"

Barriers: Promotion

Description: Problems being promoted; lack of criteria for promotion; arbitrary promotion decisions; anything that casts problems with promotion as a potential barrier. References to glass ceiling go here. Examples: "When I look up there are just not that many spots," "How they decide who gets promoted isn't really clear to everyone"

Barriers: Respect/Men Taking Control

Description: Feeling not listened to or not treated with consideration or respect. Also includes the desire for more autonomy (e.g., signing yourself in or out of work, setting own schedule, as long as it is identified as a problem). Examples: "Some of the men overpower and step in and compensate," "I had enough of not being recognized."

Barriers: Women's Characteristics

Description: Failure to communicate effectively; personality traits; anything that puts blame on women; self-doubt. Examples: "If I liked girly girl stuff" "If they may be teary-eyed, then its frowned upon by the male peers"

Cultural Norms

Description: Informal knowledge that shapes the employees behavior; how things are done. Examples: "you're just gonna put in 45 hours" "IS gets paid at that rate"

Gender Differences

Description: Express discussion of there being a gender difference - maybe in how they are treated, interest areas they like, different emotions, and the way they deal with stress. Direct comparisons of men and women. Examples: "This girl came to my department. The guy in front of her was salaried. She was hourly," "Women kind of have a different mindset than men do"

Information Systems (IS)

Description: Reference to job, getting job done, special programs, projects, job duties; characteristic of IT; move into IT. Examples: "If you are developing systems," "We are in the middle of a networking project"

Job Qualities

Description: Positive job-related qualities, positively viewed actions, qualities needed, skill sets, positive qualities in someone; relationship with someone where feel comfortable or positive. Examples: "I can say a lot of good things for the people who are willing to go those extra hours to meet that customer demand," "I learned the system and hit the ground running"

Lack of Control

Description: This is how things are, this is what I have seen, cannot change it, take what you can get; situation described as out of control, just coincidence; denials and disclaimers. Examples: "I don't know who is getting paid what," "You can't change it."

Managing Family Responsibilities

Description: Women do the most of the child rearing. Include here unless identified as why a woman left a job or if talk about the choice to have a family being a decision they had made. Examples: "It's kind of expected that the Mom will take care of things," "If the family is more important, which it should be"

Men

Description: Male activities and interests; things men can do that married women or mothers cannot. Examples: "They tend to play golf and baseball together" "It's easy for them to work 'til 7:00 every night"

My Situation is Different

Description: Says their situation is different or that they cannot relate to what is being discussed. Generally they will say "my situation is different". Also, this includes when they mention a situation facing a specific individual or mention specific people when the comment can't be coded any other way. Examples: "She had different circumstances," "My users don't tend to, they may try, but they don't push me around that much"

Negative Comments

Description: Statements regarding negative emotions such as: beat down, irritated; negative situations, dilemma. Examples: "turmoil creates demoralization" "we weren't happy"

Opportunities

Description: New, growth or promotion opportunities; broaden horizons; challenging job. Examples: "Our job is very challenging," "We don't do the same thing two days in a row"

Organization

Description: References to company size, changes, number of buildings, or other companies / comparisons. Examples: "the whole matrix organization" "a lot of change in the organization"

Positive Comments

Description: Statements regarding positive emotions such as: happy, love, interesting, like what I do; feel better. Examples: "getting it done," "we were successful," "they were a lot happier."

Result of Own Decision

Description: Make decisions that move their career from being their primary interest; people are responsible for their own decisions; people show different levels of initiative; includes statements of decision to remain in organization / position; statements indicating they take control of their career. Examples: "I think everybody who has children makes that decision," "I personally have made that decision [to put family first]"

Time

Description: Mentions of time either as problems or simply hours worked. Examples: "Now you have to work just 6:30am until 3:30pm or 7:00am to 4:00pm"

Turnover

Description: Statements about leaving an organization. Examples: "A very good employee that was going to leave the company," "Otherwise they just come in, get the training, and go"

Turnover: Family

Description: Left organization or IS due to family considerations like change in family status or family pressure. Examples: "A few women have left when they've had their second child"

Turnover: Job Characteristics

Description: Comments indicating an employee left a job because she was either bored, stressed out, overloaded, not acquiring needed new skills, job or projects were unclear, worked long hours, or had too much challenge on the job. Examples: "If you are stuck in old technology," "I was on call 24 hours a day 7 days a week"

Turnover: Management

Description: Left organization or IS group because management was inattentive, unfair, ineffective or inappropriate. Examples: "The good ol' boys thing kind of shoved her out of IT," "I was tired of working for people who were sitting around waiting to retire."

Work Schedule Flexibility

Description: Positive ability to have flexible work schedules. Positive statements regarding breaking rules, asking for flexibility, flexibility exists, autonomy, personal business, part of give and take, or choice. Examples: "If they have personal things that they need to leave and do, then they go do it," "I wanted to work part-time and still get some pay during that time"

Work Stress

Description: Direct statements regarding stress or descriptions of elements within the working environment that are stressful. May include long work hours, problems with competing demands, things moving at Internet speed or competition. Examples: "I'm definitely over committed," "I couldn't take any more change right now," "It's a competition to move up."